Svistak Vladimir

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Education

Chemical Engineering Center, ITMO University 2023–2027

Experience

Practice in the Laboratory of Physical Chemistry November 2023 – December 2024

ITMO faculty of Infochemistry

- Growing microgreens using hydroponics
- Preparation of the nutrient medium
- Dependency injection with Hilt
- Synthesis of melamine cyanurate particles
- Work in sterile conditions
- Working with a fluorescence microscope

Laboratory of Molecular Biology of Fungi November 2024 – January 2025

Komarov Botanical Institute

- Isolation of fungal DNA by the magnet method
- Preparation of mushroom samples for DNA isolation
- Compilation of phylogenetic trees using the maximum likelihood method
- Alignment by various BLAST algorithms
- Working with FastQC

ML prediction of aptamer–small molecule interactions August 2025 – Present

ITMO faculty of Chemistry and AI

- Creating artificial aptamers for training (random assembly, cVAE, GRU)
- Using contrastive learning to predict the interaction between aptamers and molecules
- Testing interaction using docking, Boltz, and other tools

Analysis of amino acid substitutions on the stability of amyloid proteins December 2025 – Present

Laboratory of Amyloids of St. Petersburg State University

- Writing specialized software to create gatekeeper mutations
- Docking proteins to test for adhesion
- Using ML/AI architectures to create a predictive model of amyloidogenic regions

CV detection of construction workers' helmets and safety vests May 2026 – Present

ML Laboratory at the Financial Institute

- Using YOLO models for object detection
- Development of a photo submission pipeline and subsequent processing / uploading to the cloud

Publications

Eremeyeva M., Svistak V. Evaluating Negative Construction for Aptamer-Small Molecule Interaction Learning. AIDaR @ NeurIPS 2026, poster.

Technical Skills

Programming languages: Python, SQL

Frameworks and libraries: Pandas, NumPy, Scikit-learn, LightGBM, RDKit, Morgan fingerprints, Seaborn, BioPython, Matplotlib, OpenCV, BeautifulSoup, requests, selenium, PyTorch, pdbfixer, MDAnalysis, freesasa, py3Dmol, OpenMM, GROMACS, AutoDock Vina

Structure prediction: AlphaFold 3, Boltz 2, RoseTTAFold, ProteinMPNN, ESM-2

ML architectures: Diffusion models, Transformer models, contrastive learning, GRU

Dev tools: Linux, Git, Docker, MLflow, Nextflow, Galaxy, BEDtools, Seqtools, SAMtools, FastQC, FoldX

Courses

Summer lecture on Bioinformatics, Institute of Bioinformatics July 2024

- NGS sequencing and data analysis
- Fundamentals of molecular modeling
- Bioinformatics in oncology

School of Bioinformatics, Center for Chemical Engineering May 2024

- Molecular docking
- Alignment and sequence analysis
- Parsing and reading information from various biological databases

Projects

CodonAdaptation

- Codon optimization
- Machine learning
- Molecular biology

Biokhakaton 2025

- Amyloids
- Amino acid substitutions
- Structural biology

DataCon 2025

- Alzheimer's therapy via the adenosine A2A receptor
- Morgan fingerprints and ligand-activity models
- De novo generation and virtual screening

Generation of AmpC β-lactamase inhibitors

- Reinforcement learning with surrogate ML models
- Scaffold split, LightGBM
- Filtering and AutoDock Vina